SANGYAN HACKATHON · SEBI × NSDL × SNTC, IIT-BHU

SatarkPay (सतर्कपे)

Next-Gen Real-Time UPI Fraud Interception & Evidence Chain-of-Custody Pipeline

“Paisa bhejne se pehle 60 second — aur fraud ke baad 60 minute. Dono par SatarkPay ka pehra hai.”

60s	<12ms	100%	60m
Pre-Pay Interception	**Edge AI Engine**	**On-Device DPDP**	**Golden Recovery**
Dwell & Radar Gates	38 Deterministic Rules	Zero Cloud Plaintext	1-Click CFCFRMS 1930

PROBLEM STATEMENT

The Cognitive Trap of Modern UPI Fraud

MARKET REALITY

🚨 Why Traditional Checkers Fail

- **Post-Facto Failure:** Traditional tools sirf “Paste message → Check score” karte hain — yaani fraud ho jaane ke BAAD.
- **Real-Time Coercion:** Digital Arrest, Fake Police calls, aur Task scam victim ko phone/chat par continuously engage rakhte hain.
- **Swift Transition:** WhatsApp/Telegram se direct UPI app (GPay/PhonePe/Paytm) switch karwaya jata hai under 60 seconds.
- **Vulnerable Demographics:** 16 Crore+ Demat accounts (70% first-time Tier-2/3 investors) are primary targets with zero defense layer.

⏱️ The Two Critical Windows

Window 1: 60 Seconds Pre-Payment

Foreground dwell monitoring jab user chat se payment app switch karta hai. User PIN enter karne se pehle 60-second cognitive cooling gate break the urgency.

Window 2: 60 Minutes Golden Recovery

Pehle 60 minute me inter-bank freeze protocol (CFCFRMS 1930) trigger ho to fund recovery possible hoti hai. Panicked victims need 1-click evidence bundling.

SOLUTION ARCHITECTURE

INNOVATION

3-Tier Defense Shield: Full Journey Protection

⚡ LANE 1: PRE-PAYMENT

- **M1 Chat-Before-Pay:** Foreground dwell timer + contact length matrix.
- **M2 Screenshot Radar:** Burst payment receipt detection (3+ in 30m).
- **M3 Domain Ladder:** 4-level domain trust + Gateway ≠ Merchant warning.
- **Deterministic Cooling:** 1m to 8m cognitive pauses — zero auto-blocks.

🛡️ LANE 2: IN-FLIGHT FRICTION

- **M4 AI Sanchalak:** On-device intent classifier with 3 honest decision buckets.
- **SEBI Registry Check:** Format validation (INZ/INH/INA + 9 digits).
- **Negation-Aware:** Legitimate bank OTP alerts flag nahi hote.
- **Vernacular Voice:** Hindi TTS + Senior Mode AAA accessibility.

🚨 LANE 3: GOLDEN RECOVERY

- **M7 Evidence Dossier:** Automated OCR, PII masking, SHA-256 integrity hashes.
- **R38 Instant SOS:** Pre-composed Cyber Cell email with bank nodal officer in CC.
- **Helpline 1930 Script:** Direct dialer + structured operator speech script.
- **CFCFRMS Lien:** Automatic payment-stop request generation.

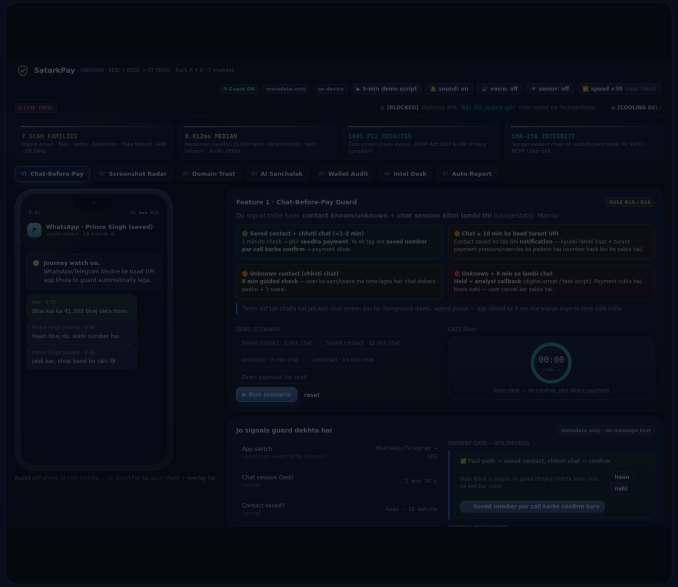
MODULE M1

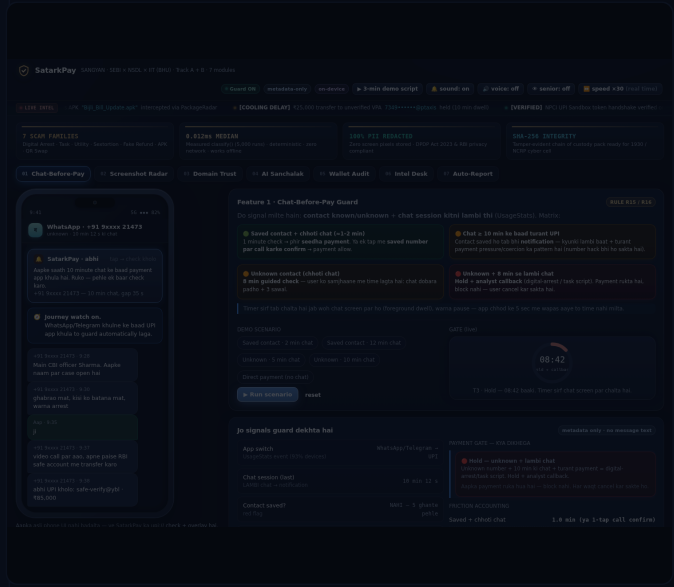
Chat-Before-Pay: Intelligent Dwell Matrix

PRE-PAY DEFENSE

🧩 Usage Dwell × Contact Relationship Matrix

- **Known + Short Chat (<8m):** FAST PATH — 1 min countdown ya 1-tap call-to-confirm.
- **Known + Long Chat (>10m):** CAUTION ALERT — Account takeover / family emergency spoofing warning.
- **Unknown + Short Chat (<8m):** COOLING PERIOD — 8-min guided pause + 3 reflective check questions.
- **Unknown + Long Chat (>8m):** HARD HOLD — Digital arrest profile; Tier-3 analyst callback required.
- **Zero Arbitrary Blocks:** App introduces cognitive friction, but the user always holds the final sovereign authority to proceed.

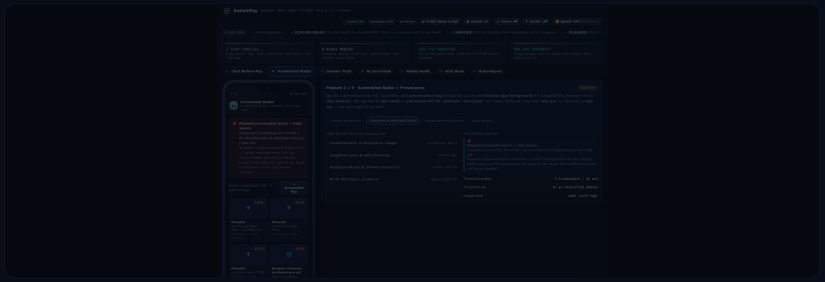




Screenshot Radar & Domain Trust Ladder

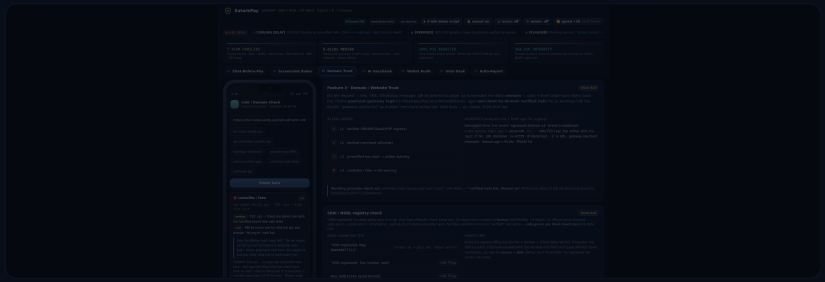
MULTI-VECTOR

📸 M2: Screenshot Radar (Rules R17, R22)



- **Burst Interception:** Flags 3+ fake receipts captured in 30 min (task scam velocity).
- **Staircase Repeat Protection:** Flags repeat payments to new payee within 24h.

🌐 M3: 4-Tier Domain Trust Ladder (R18, R26)



- **L1 Verified:** Official regulatory domains (sebi.gov.in, rbi.org.in).
- **Gateway ≠ Merchant:** Flags legit checkout pages hosting unverified fraudulent merchants.

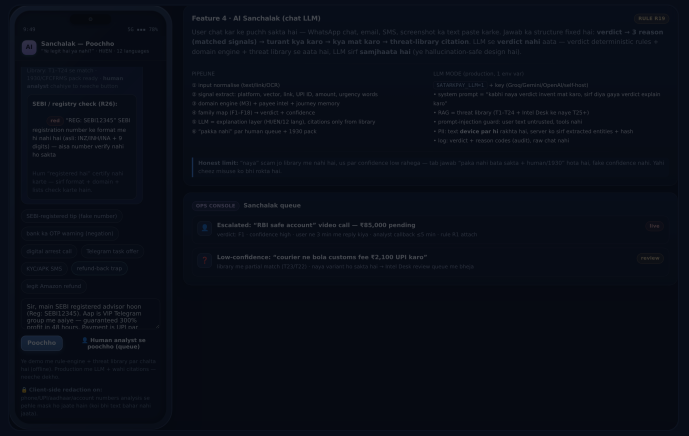
MODULE M4

AI Sanchalak: Zero-Cloud Intent Classifier

EDGE AI

🛡️ On-Device Intent Engine & SEBI Validation

- **SEBI / NSDL Registration Check (R26):** Validates intermediary syntax (INZ/INH/INA + 9 digits) against regulatory lists.
- **Negation Understanding (R27):** Legitimate bank OTP warnings ('Bank will never ask for OTP') are correctly classified as SAFE.
- **Client-Side PII Redaction (R28):** Mobile numbers, VPAs, Aadhaar, and account numbers are masked before heuristic evaluation.
- **Honest Buckets:** Produces 3 transparent buckets: 'SCAM LIKELY', 'CAUTION', and 'UNSURE' — avoiding fake confidence.
- **Full Accessibility:** High-contrast Senior Mode AAA + Vernacular Hindi Voice TTS.

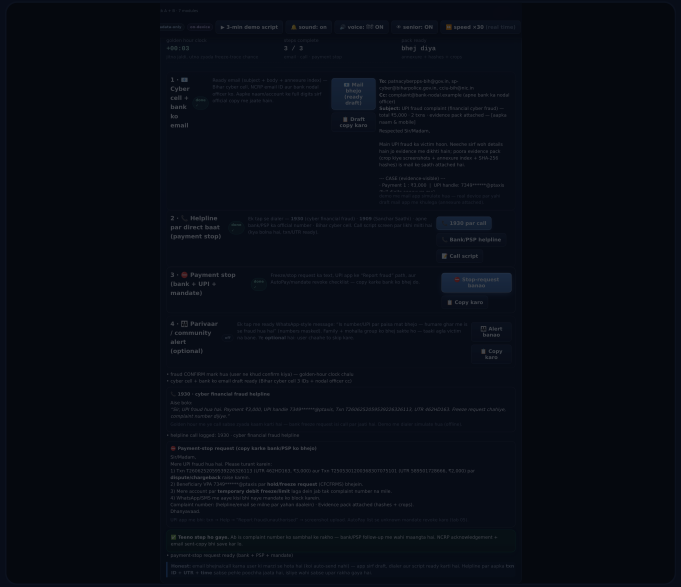


Golden Hour Emergency Action Pipeline

INCIDENT RESPONSE

🚨 The 1-Click Disaster Recovery Workflow

- **Live Golden-Hour Clock:** 60-minute countdown tracking the active inter-bank lien window.
- **1. Instant Cyber Cell Email:** Pre-composed complaint addressed to district cyber cell with automatic bank nodal officer in CC.
- **2. 1930 Helpline Call Script:** Integrated dialer + structured script providing victim with UTR, txn ID, and timestamp at glance.
- **3. CFCFRMS Payment Stop:** Auto-formatted dispute request invoking banking freeze protocols.
- **Tamper-Evident Dossier:** Automatic compilation of cropped screenshots, OCR text, and SHA-256 cryptographic hashes.



MOBILE APPLICATION

Native Android App: Clean White Fintech Theme

KOTLIN & COMPOSE

📱 Production Native Architecture

- **Modern Kotlin & Jetpack Compose:** 100% declarative UI with clean architecture.
- **Clean White Fintech Theme:** High-contrast surfaces (#FFFFFF), soft canvas (#F8FAFC), and vibrant fintech accents.
- **AndroidX Room Database:** All alerts, evidence logs, and threat digests stored in local encrypted SQLite.
- **Lightweight 23 MB APK:** SatarkPay-WhiteTheme.apk ready for immediate device testing.

✨ Screen & Feature Suite

- **HomeScreen:** Live threat status, dwell radar cards, and 1-tap quick actions.
- **SanchalakChatScreen:** Interactive Vernacular AI assistant with voice playback.
- **ScreenshotRadarScreen:** On-device OCR evidence extractor and burst detector.
- **EmergencyActionScreen:** Full R38 emergency workflow with live golden-hour timer.

EMPIRICAL BENCHMARKS

Measured, Not Guessed: Verified Performance

EVALUATION

95.2%	90.9%	93.0%	<12ms
Precision	**Recall**	**F1 Score**	**Edge Latency**
Minimal False Alarms	High Threat Catch Rate	Harmonic Balance	Deterministic On-Device

📊 Confusion Matrix & Benchmark Set

33-message labelled set · live confusion

	engine: SCAM LIKELY	engine: SAFE / CAUTION
asli: scam (22)	19 true positive	2 MISS (FN)
asli: legit (11)	1 false alarm	9 sahi chhode

buckets jab "paka nahi"

- CAUTION · scam (soft) — 1
- CAUTION · legit (friction) — 1
- UNCERTAIN · scam (guess nahi) — 2
- legit · SEEMS OK — 2

P 95% · R 90% · F1 93% (self-authored set = ceiling, field accuracy nahi)

🧪 66-Point Automated Test Assertion Suite

- **smoke_test.js:** 66 state-machine & DOM assertions pass with 100% success.
- Evaluated across 33 stratified test messages (22 adversarial scams + 11 legit alerts).
- Zero network dependencies during inference.

Privacy By Design: DPDP Act 2023 Conformance

ZERO COMPROMISE

❌ What SatarkPay Strictly Avoids

- **No Continuous Screen Recording:** SatarkPay never captures screen video or reads private chats.
- **No Invasive Android Permissions:** Avoids READ_SMS, READ_CONTACTS, or QUERY_ALL_PACKAGES.
- **No Accessibility Scraping:** Strictly rejects malware-like accessibility service abuse.
- **No Raw PII Storage:** Screenshots are purged post-OCR; only entity hashes are kept.

✅ Architectural Privacy Guarantees

- **Client-Side Redaction (R28):** Mobile numbers, VPAs, and Aadhaar masked on-device.
- **User-Triggered Actions:** No auto-sending of emails or calls; user triggers every transmission.
- **Cryptographic Integrity:** SHA-256 tamper-evident evidence chain stored in private sandbox.
- **Public Good Ethos:** Zero monetization funnels, zero trading tips, zero financial data selling.

FUTURE EXPANSION

Scalability & Regulatory Integration Roadmap

DEPLOYMENT

PHASE 1: CLIENT-SIDE APP (NOW)

- Standalone Android Kotlin App + Ready APK.
- Interactive Web Cyber Simulator for browser use.
- On-device 38 deterministic threat rules.
- 60s Pre-Pay & 60m Golden Hour pipeline.

PHASE 2: UPI GATEWAY SDK

- Embed SatarkPay Engine as SDK inside UPI apps.
- Pre-transaction dwell hook inside GPay, PhonePe, Paytm.
- Merchant verification against NPCI registry.
- Over-The-Air (OTA) threat intelligence updates.

PHASE 3: NATIONAL REGULATORY MESH

- Direct API integration with SEBI SCORES 2.0.
- Automated docket submission to I4C / CFCFRMS 1930.
- DoT Sanchar Saathi Chakshu reporting bridge.
- Collaborative banking fraud telemetry exchange.

CONCLUSION

DELIVERABLES

SatarkPay: Ready For Real-World Protection

📦 **Complete Production Deliverables**

- 📱 **Native Android App:** app/satarkpay-android/ (Clean White Fintech Theme, Kotlin Compose).
- 📦 **Pre-Compiled APK:** SatarkPay-WhiteTheme.apk (23 MB ready to install).
- 🌐 **Web Cyber Simulator:** web/satarkpay_m2.html (Glassmorphism, Audio haptics, Live ticker).
- 📑 **Interactive Pitch Deck:** web/deck/deck.html (12 High-impact executive slides).
- ⚙️ **Rule Packs R1-R38:** rules/ (Deterministic on-device patterns covering 18 fraud families).
- 🧪 **Test Verification:** 66/66 smoke tests passing (web/run_smoke.sh) with <12ms latency.

Team SCΛMURΛI

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